

Safety Data Sheet(SDS)

1. Identification of the substance/mixture and of the company/undertaking

1) Product identifier : 2-ETHYL HEXYLACRYLATE (2-EHA)

2) Relevant identified uses of the substance or mixture and uses advised against

○ Relevant identified uses

Adhesives, sealants, Coatings and paints, thinners, paint removers, Polymer preparations and compounds

○ Uses advised against

Prohibited from use other than recommended

3) Supplier information

○ Company name [Manufacture]

Company : LG Chem, Ltd.

Address : 58, Yeosusandan 4-ro, Yeosu-si, Jeollanam-do, Republic of Korea

Emergency number : 080-880-0454

2. HAZARD IDENTIFICATION

1) Hazard classification

- Flammable liquids Category 4
- Skin corrosion/irritation Category 2
- Skin sensitization Category 1
- Carcinogenicity Category 2
- Specific target organ toxicity single exposure Category 3(Respiratory tract irritation)
- Hazardous to the aquatic environment, long-term (chronic) Chronic 3

2) Allocation label elements

Hazard pictograms



Signal word

- WARNING

Hazard statements

- H227 Combustible liquid
- H315 Causes skin irritation
- H317 May cause an allergic skin reaction
- H335 May cause respiratory irritation
- H351 Suspected of causing cancer
- H412 Harmful to aquatic life with long lasting effects

Precautionary statements

- Prevention

- P201 Obtain special instructions before use.
- P202 Do not handle until all safety precautions have been read and understood.
- P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
- P261 Avoid breathing mist/vapours/spray.
- P264 Wash skin thoroughly after handling.
- P271 Use only outdoors or in a wellventilated area.
- P272 Contaminated work clothing should not be allowed out of the workplace.
- P273 Avoid release to the environment.
- P280 Wear protective gloves/protective clothing/eye protection/face protection.

- Response

- P302+P352 If you get on your skin: Wash with a large amount of water.
- P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
- P308+P313 If exposed or concerned: Get medical advice/attention.
- P312 If you feel uncomfortable, receive medical institutions and doctors' consultation.
- P321 When contacting with substances, make a treatment such as rinsing the skin and eyes in the water flowing more than 20 minutes.
- P332+P313 If skin irritation occurs: Get medical advice/attention.
- P333+P313 If skin irritation or a rash occurs: Get medical advice/attention.
- P362+P364 Take off contaminated clothing and wash it before reuse.
- P370+P378 In case of fire: Use carbon dioxide, Water, Alcoholic to extinguish.

- Storage

- P403 Store in a wellventilated place.
- P403+P233 Store in a wellventilated place. Keep container tightly closed.
- P405 Store locked up.

- Disposal

- P501 Dispose of contents and containers according to the legislation of the waste

3) Other hazards

- No data available

○ Product NFPA Level

Health	Flammability	Reactivity
2	2	2

(※ 0 = Stable , 1 = Low , 2 = Medium , 3 = High , 4 = Very High)

3. Composition/Information on ingredients

Components	Common name	CAS No.	PCT(wt%)
2-Ethylhexyl acrylate	2-Ethylhexyl acrylate	103-11-7	100

4. FIRST AID MEASURES

1) Following eye contact

- Seek immediate medical assistance.
- In case of contact with substance, immediately flush skin or eyes with running water for at least 20 minutes.

2) Following skin contact

- For minor skin contact, avoid spreading material on unaffected skin.
- Seek immediate medical assistance.
- For hot product, immediately immerse in or flush the affected area with large amounts of cold water to dissipate heat.
- In case of burns, immediately cool the affected area with cold water for as long as possible and do not remove clothing that is stuck to the skin.
- In case of contact with substance, immediately flush skin or eyes with running water for at least 20 minutes.
- Wash skin with soap and water.
- Remove and isolate contaminated clothing and shoes.

3) Following inhalation

- Do not perform artificial respiration by mouth-to-mouth method, but use appropriate respiratory medical equipment.
- Seek immediate medical assistance.
- Keep victim warm and quiet.
- Move to fresh air.
- Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.
- Administer oxygen if breathing is difficult.
- Give artificial respiration if victim is not breathing.

4) Following ingestion

- Do not perform artificial respiration by mouth-to-mouth method, but use appropriate respiratory medical equipment.
- Seek immediate medical assistance.

5) Advice to physician

- Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.

- Exposures require specialized first aid with contact and medical follow-up .

5. FIRE FIGHTING MEASURES

1) Suitable (and unsuitable) extinguishing media

- Suitable extinguishing media
 - CO₂.
 - Dry chemical.
 - For mixtures containing alcohol or polar solvent: Alcohol-resistant foam.
 - Water spray.
 - Use alcohol foam, carbon dioxide, or water spray when fighting fires involving this material.
 - Regular foam.
 - Use dry sand or earth to smother fire.
- Unsuitable extinguishing media
 - High-pressure water.
 - Direct water.

2) Special hazards arising from the substance or mixture

- Pyrolytic product
 - Can decompose at high temperatures forming toxic gases.
 - During burning, irritating and highly toxic gases may be produced by thermal decomposition or combustion.
- Risk of fire and explosion
 - Containers may explode when heated.
 - When heated, vapors may form explosive mixtures with air: indoors, outdoors and sewers explosion hazards.
 - May violently polymerize and result in fire and explosion.
 - HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
 - Runoff may create fire or explosion hazard.
 - Vapor explosion hazard indoors, outdoors or in sewers.
 - Can form explosive mixtures at temperatures at or above the flashpoint.
 - Vapors may form explosive mixtures with air.
 - Vapors may travel to source of ignition and flash back.
- Other
 - Some liquids produce vapors that may cause dizziness or suffocation.

- Contact may cause burns to skin and eyes.
- Fire may produce irritating and/or toxic gases.

3) Special protective equipment for firefighters

- Rescuers should put on appropriate protective gear.
- Cautions ; Most of liquids are lighter than water.
- Most vapors are heavier than air. They will spread along ground and collect in low or confined areas (sewers, basements, tanks).
- Dike fire-control water for later disposal; do not scatter the material.
- Move containers from fire area if you can do it without risk.
- Evacuate area and fight fire from a safe distance.
- Fire involving Tanks: For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible, withdraw from area and let fire burn.
- Fire involving Tanks: Cool containers with flooding quantities of water until well after fire is out.
- Fire involving Tanks: Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank.
- Fire involving Tanks: Fight fire from maximum distance or use unmanned hose holders or monitor nozzles.
- Fire involving Tanks: ALWAYS stay away from tanks engulfed in fire.

6. ACCIDENTAL RELEASE MEASURES

1) Health considerations and protective equipment

- Do not touch or go near exposed material.
- The very fine particles can cause a fire or explosion, eliminate all ignition sources.
- ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area).
- All equipment used when handling the product must be grounded.
- Clean up spills immediately, observing precautions in Protective Equipment section.
- Stop leak if you can do it without risk.
- Do not touch damaged containers or spilled material unless wearing appropriate protective clothing.
- A vapor suppressing foam may be used to reduce vapors.
- Cover with plastic sheet to prevent spreading.
- Please note that materials and conditions to be avoided.

2) Environmental precautions

- Runoff may cause pollution.
- Prevent entry into waterways, sewers, basements or confined areas.
- Keep out of waterways.

3) For cleaning up

- Absorb or cover with dry earth, sand or other non-combustible material and transfer to containers.
- Large Spill: Dike far ahead of liquid spill for later disposal.
- Absorb spill with inert material (e.g., dry sand or earth), then place in a chemical waste container.
- Small Spill: Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal.

- Dike and collect water used to fight fire.
- Absorb the liquid and scrub the area with detergent and water.
- Use clean non-sparking tools to collect absorbed material.

7. HANDLING AND STORAGE

1) Precautions for safe handling

- Avoid breathing vapors from heated material.
- Loosen closure cautiously before opening.
- CAUTION: High temperature.
- Handling refer to engineering control/personal protection section.
- All equipment used when handling the product must be grounded.
- DO NOT PRESSURIZE, CUT, WELD, BRAZE, SOLDER, DRILL, GRIND, OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES OF IGNITION.
- Caution: Heat.
- Follow all SDS/label precautions even after container is emptied because they may retain product residues.
- Avoid prolonged or repeated contact with skin.
- Measure atmospheric oxygen concentration and ventilate the area during the operation since low-closed area can cause oxygen deficiency.
- Do not enter storage area unless adequately ventilated.
- Use care in handling/storage.
- Please note that materials and conditions to be avoided.
- Use only in a well-ventilated area.

2) Conditions for safe storage (including any incompatibilities)

- Empty drums should be completely drained, properly bunged, and promptly returned to a drum reconditioner, or properly disposed of.

8. EXPOSURE CONTROLS AND PERSONAL PROTECTION

1) Chemical exposure limits, Biological exposure standard

Components	Occupational exposure limits	ACGIH	Biological standard
2-Ethylhexyl acrylate	TWA : Not applicable STEL : Not applicable	TWA : Not applicable STEL : Not applicable	Not applicable

2) Appropriate engineering controls

- Use process enclosures, local exhaust ventilation, or other engineering controls to control airborne levels below recommended exposure limits.
- If user operations generate dust, fume, or mist, use ventilation to keep exposure to airborne contaminants below the exposure limit.
- Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

3) Personal protection equipment

- Respiratory protection
 - In case of insufficient oxygen (<19.6%), wear a supplied air mask or self-contained respirator.

- Wear breathing protection, which needs a confirmation from the Korea Occupational Safety and Health Agency.
- Eye protection
 - Wear suitable protective goggles and face shields.
- Hand protection
 - Wear suitable protective gloves.
- Body protection
 - Wear suitable protective clothing.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	transparent liquid
Physical state	Liquid
Colour	colorless
Odour	Characteristic odour
Odour threshold	0.02 mg/l
pH	No data available
Melting point/freezing point	-90 °C
Initial boiling point and boiling range	213.5°C
Flash point	86°C
Evaporation rate	No data available
Flammability(solid, gas)	Flammable liquid
Upper/lower flammability or explosive limits	6.4 / 0.8 %
Vapour pressure	0.178 mmHg (@ 25 °C)
Solubility(ies)	100 mg/l (@ 25 °C)
Vapour density	6.35
Relative density	0.89
n-octanol/water partition coefficient	3.67
Auto ignition temperature	252 °C
Decomposition temperature	No data available
Viscosity	1.38mm ² /s
Molecular weight(mass)	184.28

10. STABILITY AND REACTIVITY

1) Stability and hazardous reactivity

- Containers may explode when heated.
- When heated, vapors may form explosive mixtures with air: indoors, outdoors and sewers explosion hazards.
- May violently polymerize and result in fire and explosion.
- Can decompose at high temperatures forming toxic gases.
- HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
- Runoff may create fire or explosion hazard.
- Vapor explosion hazard indoors, outdoors or in sewers.
- Can form explosive mixtures at temperatures at or above the flashpoint.
- Some liquids produce vapors that may cause dizziness or suffocation.
- Contact may cause burns to skin and eyes.
- Vapors may form explosive mixtures with air.
- Fire may produce irritating and/or toxic gases.
- Fire may produce irritating, corrosive and/or toxic gases.

2) Conditions to avoid

- Containers may explode when heated.
- Ignition source(heat, spark, flame, etc.).
- Heat, contamination.
- Heat.
- Some liquids produce vapors that may cause dizziness or suffocation.
- Contact may cause burns to skin and eyes.
- Fire may produce irritating and/or toxic gases.

3) Incompatible materials

- Combustibles, reducing material.
- Containers may explode when heated.
- Some liquids produce vapors that may cause dizziness or suffocation.
- Contact may cause burns to skin and eyes.
- Fire may produce irritating and/or toxic gases.

4) Hazardous decomposition products

- Corrosive/toxic fume.
- Heat.
- Irritating and/or toxic gas.
- Irritating, corrosive and/or toxic gas.
- During burning, irritating and highly toxic gases may be produced by thermal decomposition or combustion.

11. TOXICOLOGICAL INFORMATION

1) Exposure route information

- Inhalation
 - May cause respiratory irritation
- Skin Contact
 - May cause an allergic skin reaction

- Causes skin irritation
- Eye Contact
 - Not applicable
- Ingestion
 - Not applicable

2) Health hazard information

- Acute toxicity
 - Acute toxicity(Oral)
LD50 4000 ~ 6000 mg/kg Test species: Rat, Source: OECD Screening Information Data Set
 - Acute toxicity(Dermal)
LD50 >10,000 mg/kg Test species: Rabbit, Source: OECD Screening Information Data Set
 - Acute toxicity(Inhalation:Gases)
No data available
 - Acute toxicity(Inhalation:Vapours)
No data available
 - Acute toxicity(Inhalation:Dust/mist)
No data available
- Skin corrosion/irritation
Severe irritation when exposed to rabbit skin, Source: OECD Screening Information Data Set
- Serious eye damage/eye irritation
Severe eye irritation in animal testing, Source: OECD Screening Information Data Set
- Respiratory sensitization
May cause skin sensitization, Mouse, OECD TG 429, Source : Ritu Chhimwal (2017)
- Skin sensitization
 - Positive reaction in human patch test - Moderate hypersensitivity in animal test, Source: OECD Screening Information Data Set
- Carcinogenicity
3 (IARC)
- Germ cell mutagenicity
Negative results in the in vitro bacterial mutation test - Negative results in the in vivo UDS assay [OECD TG 486], chromosomal aberration mutagenicity test, Source: OECD Screening Information Data Set
- Reproductive toxicity
No data available
- Specific target organ toxicity single exposure

An acute inhalation toxicity test in rats demonstrated nasal and eye irritation, a significant local irritant potential detected on the skin, and a slight irritant potential detected on the rabbit conjunctiva., Source : BASF AG (1958)

○ Specific target organ toxicity repeated exposure

In a 90-day rat repeated toxicity test, minimal liver damage was observed at 100 ppm, with high liver enzyme activity., Source: OECD Screening Information Data Set

○ Aspiration hazard

No data available

12. ECOLOGICAL INFORMATION

1) Aquatic toxicity

- Fish

LC50 1.8 mg/l 96 hr *Oncorhynchus mykiss*, Source: OECD Screening Information Data Set

- Crustacea

EC50 1.3 mg/l 48 hr *Daphnia magna*, Source: OECD Screening Information Data Set

- Aquatic algae

EC50 44 mg/l 72 hr Other (*Scenedesmus quadricauda*), Source: International Uniform Chemical Information Database(IUCLID)

2) Persistence and degradation

- n-octanol water partition coefficient

3.67 log Kow, Source: ICSC

- Degradation

No data available

- Biodegradation

(BOD(14 days) 51.7%), Source: National Institute of Technology and Evaluation(NITE)

3) Bioaccumulative potential

280, Source: National Library of Medicine/Hazardous Substances Data Bank(NLM/HSDB)

4) Mobility in soil

54954

5) Other adverse effects

No data available

13. DISPOSAL CONSIDERATIONS

1) Disposal methods

- Recyclables must be collected in portable bins and disposed of legally according to national regulations.

2) Precautions (including disposal of contaminated container of package)

- Avoid direct spillage to rivers, lakes, soil and drains.

14. TRANSPORT INFORMATION

- 1) UN No. : 3334
 - 2) Proper shipping name : AVIATION REGULATED LIQUID, N.O.S.
 - 3) Class or division : 9
 - 4) Packing group : Not applicable
 - 5) Marine pollutant : Not applicable
 - 6) Special safety response for transportation or transportation measure :
 - Emergency measures in case of fire : Not applicable
 - Emergency measures in the effluent : Not applicable
- ADR
- Tunnel restriction code : Not applicable
- IMDG
- Marine pollutant : Not applicable
- Air transport(IATA)
- UN No. : 3334
 - Proper shipping name : AVIATION REGULATED LIQUID, N.O.S.
 - Class or division : 9
 - Packing group : Not applicable

15. REGULATORY INFORMATION

- 1) Occupational Safety and Health Act in Korea
 - Not applicable
- 2) Toxic Chemical Control Act in Korea
 - Not applicable
- 3) Safety Control of Dangerous Substances Act in Korea
 - Class 4> Third petroleum (water-insoluble)(Designated quantity:2000ℓ)
- 4) Wastes Control Act in Korea
 - Not applicable
 - In case of disposal, it must be disposed of in accordance with Article 13 of the Waste Management Act.
- 5) Other regulations in KOREA and Abroad regulations
 - ETC regulation
 - Not applicable

- Not applicable
- Act on the registration and evaluation of chemicals
Existing Commercial Chemical Substances

16. OTHER INFORMATION

1) Reference

2) Date of first issue : 1997-01-20

3) Revision date

- Revised date count : 17
- Last revised date : 2025-06-30
- Last revised history : 2-EHA 한국 롤셋 MSDS 영문버전 개정

4) Other

ORIGINAL CONTROLLED